Heart Disease Prediction using Machine Learning

Heart disease can be predicted based on various symptoms such as age, gender, heart rate, etc. and reduces the death rate of heart patients.

Due to the increasing use of technology and data collection, we can now predict heart disease using machine learning algorithms. Now let's go further with the task of heart disease prediction using machine learning with Python.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

%matplotlib inline
sns.set_style("whitegrid")
plt.style.use("fivethirtyeight")
```

```
df = pd.read_csv("heart_disease_dataset.csv")
df.head()
```

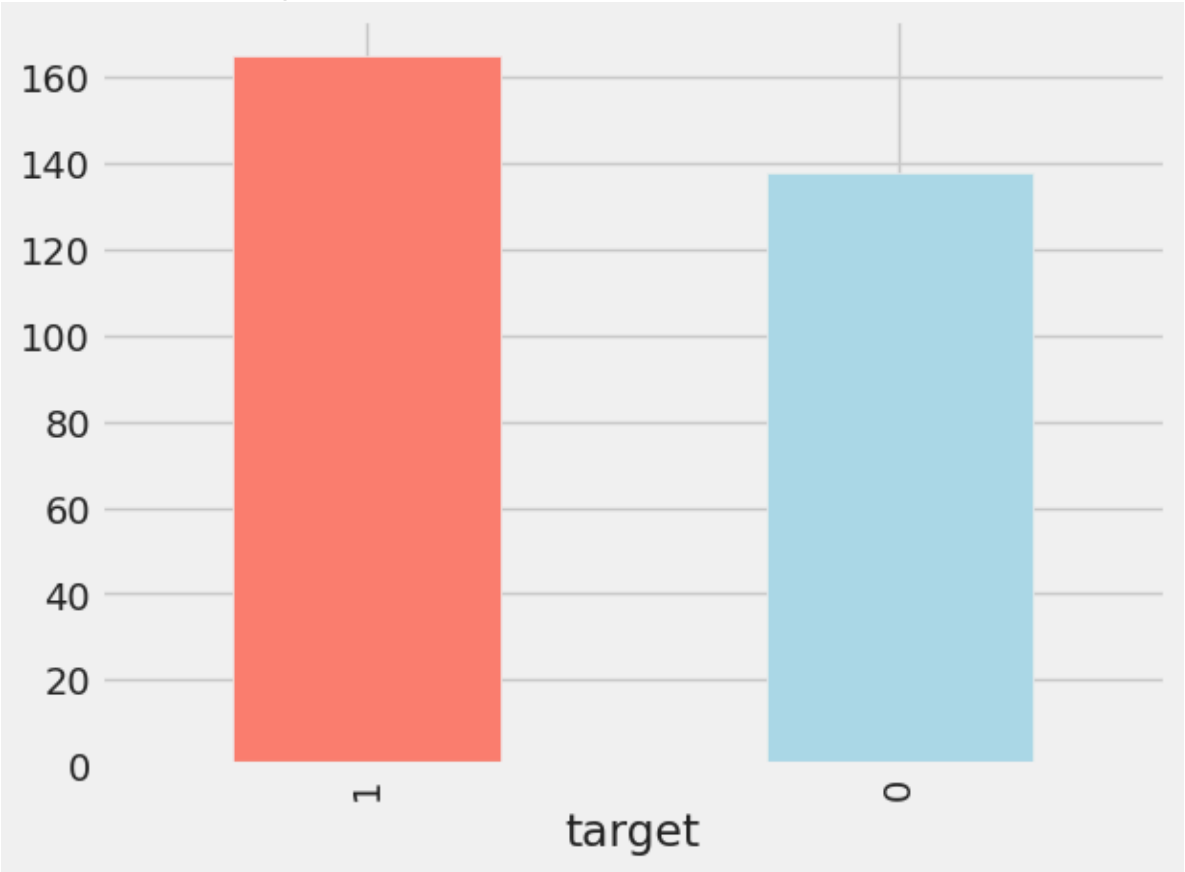
	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope
0	63	1	3	145	233	1	0	150	0	2.30	0
1	37	1	2	130	250	0	1	187	0	3.50	0
2	41	0	1	130	204	0	0	172	0	1.40	2
3	56	1	1	120	236	0	1	178	0	0.80	2
4	57	0	0	120	354	0	1	163	1	0.60	2

```
pd.set_option("display.float", "{:.2f}".format)
df.describe()
```

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exan
count	303.00	303.00	303.00	303.00	303.00	303.00	303.00	303.00	303.0
mean	54.37	0.68	0.97	131.62	246.26	0.15	0.53	149.65	0.3
std	9.08	0.47	1.03	17.54	51.83	0.36	0.53	22.91	0.4
min	29.00	0.00	0.00	94.00	126.00	0.00	0.00	71.00	0.0
25%	47.50	0.00	0.00	120.00	211.00	0.00	0.00	133.50	0.0
50%	55.00	1.00	1.00	130.00	240.00	0.00	1.00	153.00	0.0
75%	61.00	1.00	2.00	140.00	274.50	0.00	1.00	166.00	1.0
max	77.00	1.00	3.00	200.00	564.00	1.00	2.00	202.00	1.0

```
df.target.value_counts().plot(kind="bar", color=["salmon", "lightblue"])
```

```
<Axes: xlabel='target'>
```



We have 165 people with heart disease and 138 people without heart disease, so our problem is balanced.

```
# Checking for messing values
df.isna().sum()
```

	0
age	0
sex	0
cp	0
trestbps	0
chol	0
fbs	0
restecg	0
thalach	0
exang	0
oldpeak	0
slope	0
ca	0
thal	0
target	0

dtype: int64

```
# This dataset looks perfect to use as we don’t have null values.
```

```
categorical_val = []
continous_val = []
for column in df.columns:
    print('==============================')
    print(f"{column} : {df[column].unique()}")
    if len(df[column].unique()) <= 10:
        categorical_val.append(column)
    else:
        continous_val.append(column)
```

```
==============================
age : [63 37 41 56 57 44 52 54 48 49 64 58 50 66 43 69 59 42 61 40 71 51 65 5
 46 45 39 47 62 34 35 29 55 60 67 68 74 76 70 38 77]
==============================
sex : [1 0]
==============================
cp : [3 2 1 0]
==============================
trestbps : [145 130 120 140 172 150 110 135 160 105 125 142 155 104 138 128 1
 122 115 118 100 124  94 112 102 152 101 132 148 178 129 180 136 126 106
 156 170 146 117 200 165 174 192 144 123 154 114 164]
==============================
chol : [233 250 204 236 354 192 294 263 199 168 239 275 266 211 283 219 340 2
 247 234 243 302 212 175 417 197 198 177 273 213 304 232 269 360 308 245
 208 264 321 325 235 257 216 256 231 141 252 201 222 260 182 303 265 309
```

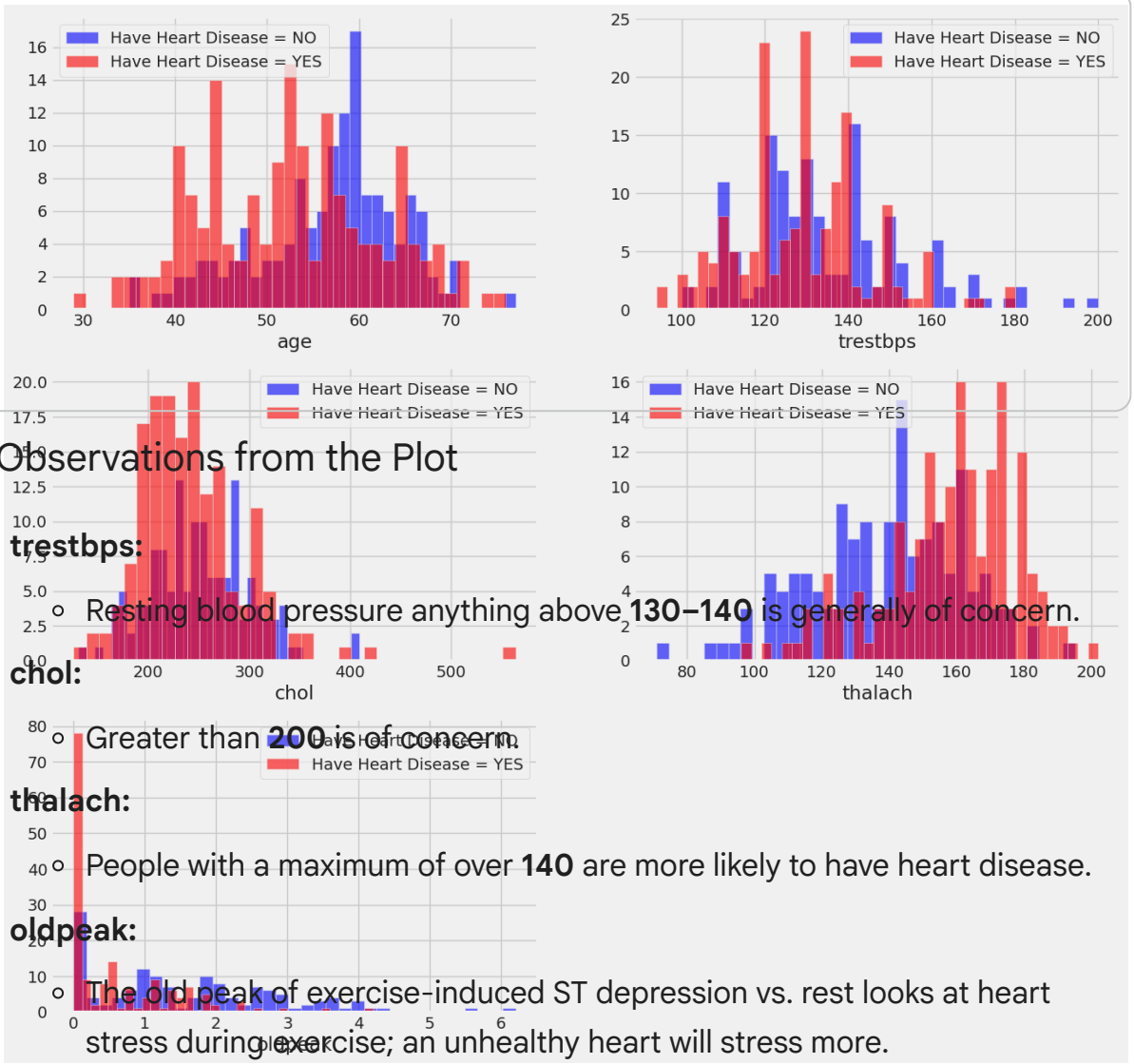
```
 186 203 183 220 209 258 227 261 221 205 240 318 298 564 277 214 248 255
 207 223 288 160 394 315 246 244 270 195 196 254 126 313 262 215 193 271
 268 267 210 295 306 178 242 180 228 149 278 253 342 157 286 229 284 224
 206 167 230 335 276 353 225 330 290 172 305 188 282 185 326 274 164 307
 249 341 407 217 174 281 289 322 299 300 293 184 409 259 200 327 237 218
 319 166 311 169 187 176 241 131]
==============================
fbs : [1 0]
==============================
restecg : [0 1 2]
==============================
thalach : [150 187 172 178 163 148 153 173 162 174 160 139 171 144 158 114 15
 179 137 157 123 152 168 140 188 125 170 165 142 180 143 182 156 115 149
 146 175 186 185 159 130 190 132 147 154 202 166 164 184 122 169 138 111
 145 194 131 133 155 167 192 121  96 126 105 181 116 108 129 120 112 128
 109 113  99 177 141 136  97 127 103 124  88 195 106  95 117  71 118 134
  90]
==============================
exang : [0 1]
==============================
oldpeak : [2.3 3.5 1.4 0.8 0.6 0.4 1.3 0.  0.5 1.6 1.2 0.2 1.8 1.  2.6 1.5 3.
 0.1 1.9 4.2 1.1 2.  0.7 0.3 0.9 3.6 3.1 3.2 2.5 2.2 2.8 3.4 6.2 4.  5.6
 2.9 2.1 3.8 4.4]
==============================
slope : [0 2 1]
==============================
ca : [0 2 1 3 4]
==============================
thal : [1 2 3 0]
==============================
target : [1 0]
```

```
plt.figure(figsize=(15, 15))

for i, column in enumerate(categorical_val, 1):
    plt.subplot(3, 3, i)
    df[df["target"] == 0][column].hist(bins=35, color='blue', label='Have He
    df[df["target"] == 1][column].hist(bins=35, color='red', label='Have Hea
    plt.legend()
    plt.xlabel(column)
```

🔍 Observations from the Plot

1. **cp (Chest pain):** People with cp = 1, 2, 3 are more likely to have heart disease compared to cp = 0.
2. **restecg (Resting EKG results):** People with restecg = 1 (abnormal heart rhythm) are more likely to have heart disease.
3. **exang (Exercise-induced angina):** People with exang = 0 (No angina during exercise) have more heart disease than those with exang = 1 (Yes).
4. **slope (Slope of ST segment during peak exercise):** People with slope = 2 (Downsloping) are more likely to have heart disease. Compared to: **0 (Upsloping) → healthier heart 1 (Flat) → minimal change, generally normal**
5. **ca (Number of major vessels (0-3) stained by fluoroscopy):** Lower value is worse → People with ca = 0 are more likely to have heart disease.
6. **thal (Thalium stress result):** People with thal = 2 (corrected defect) are more likely to have heart disease.

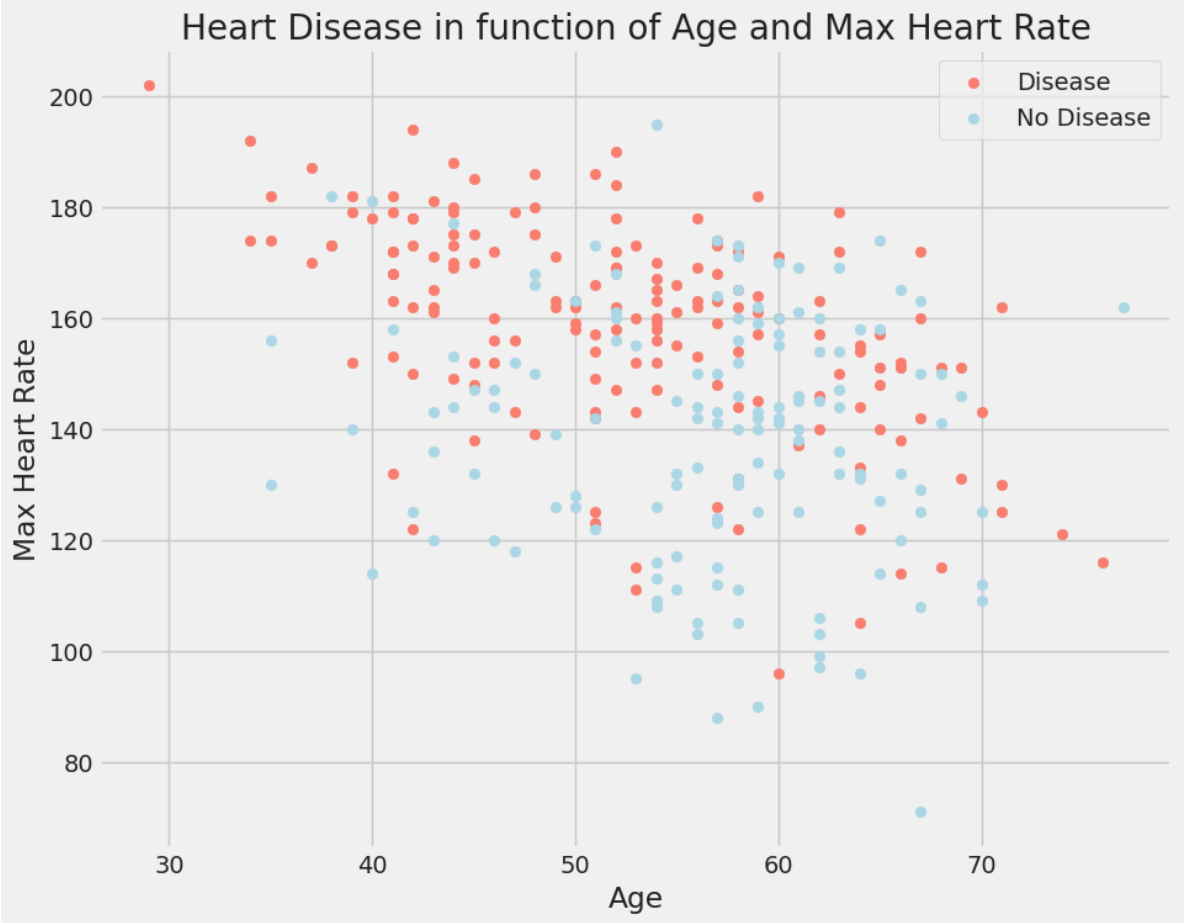
```
plt.figure(figsize=(15, 15))

for i, column in enumerate(continous_val, 1):
    plt.subplot(3, 2, i)
    df[df["target"] == 0][column].hist(bins=35, color='blue', label='Have He
    df[df["target"] == 1][column].hist(bins=35, color='red', label='Have Hea
    plt.legend()
    plt.xlabel(column)
```



Observations from the Plot

- **trestbps:**
  - Resting blood pressure anything above **130–140** is generally of concern.
- **chol:**
  - Greater than **200** is of concern.
- **thalach:**
  - People with a maximum of over **140** are more likely to have heart disease.
- **oldpeak:**
  - The old peak of exercise-induced ST depression vs. rest looks at heart stress during exercise; an unhealthy heart will stress more.

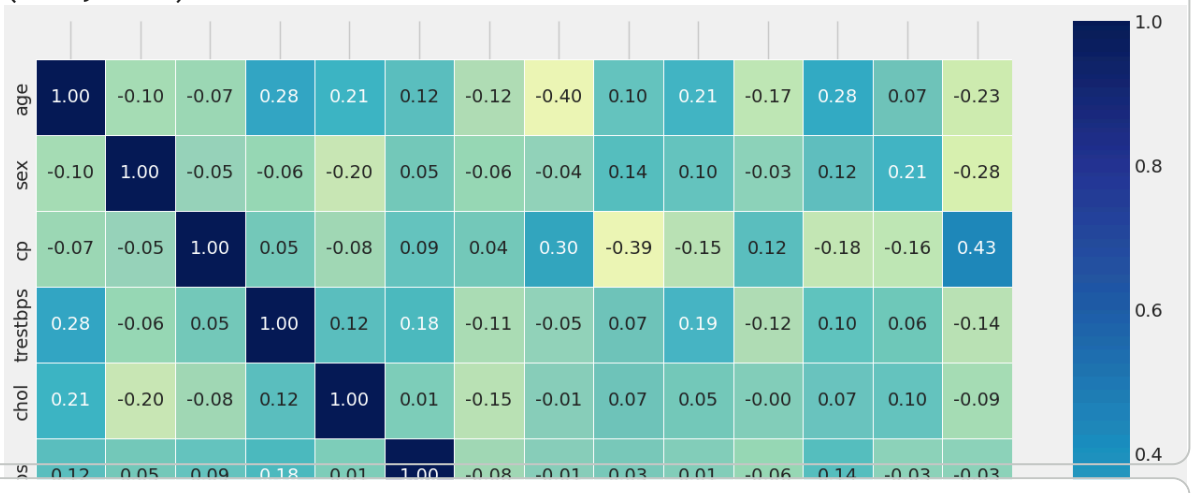
```
# Create another figure
plt.figure(figsize=(10, 8))

# Scatter with postivie examples
plt.scatter(df.age[df.target==1],
            df.thalach[df.target==1],
            c="salmon")

# Scatter with negative examples
plt.scatter(df.age[df.target==0],
            df.thalach[df.target==0],
            c="lightblue")

# Add some helpful info
plt.title("Heart Disease in function of Age and Max Heart Rate")
plt.xlabel("Age")
plt.ylabel("Max Heart Rate")
plt.legend(["Disease", "No Disease"]);
```



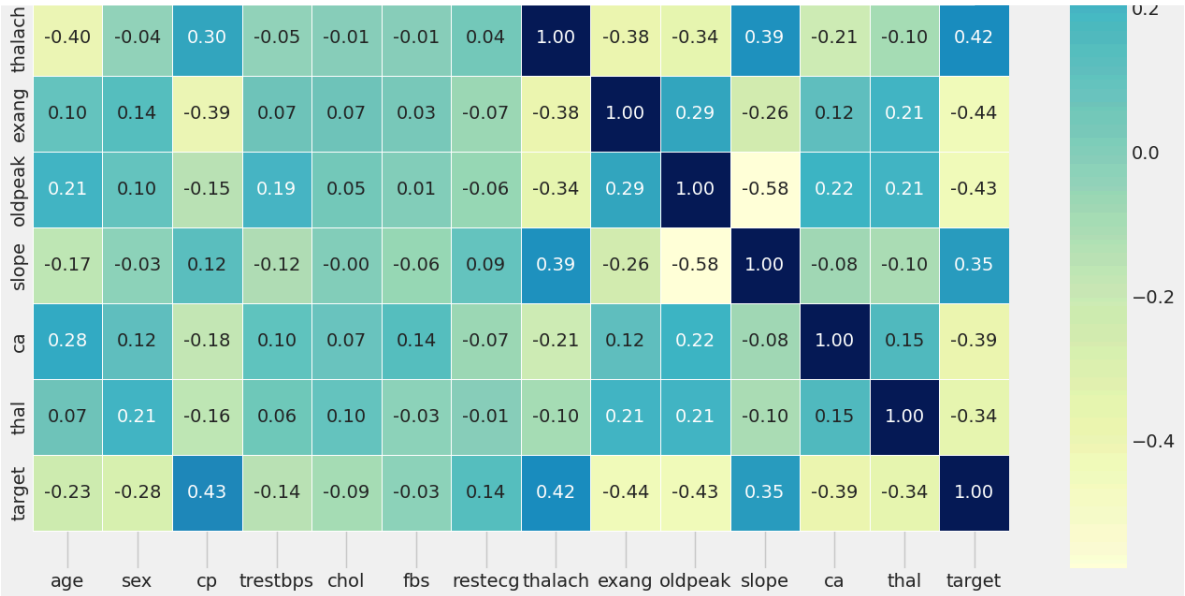
Correlation Matrix

```
# Let's make our correlation matrix a little prettier
corr_matrix = df.corr()
fig, ax = plt.subplots(figsize=(15, 15))
ax = sns.heatmap(corr_matrix,
                 annot=True,
                 linewidths=0.5,
                 fmt=".2f",
                 cmap="YlGnBu");
bottom, top = ax.get_ylim()
ax.set_ylim(bottom + 0.5, top - 0.5)
```

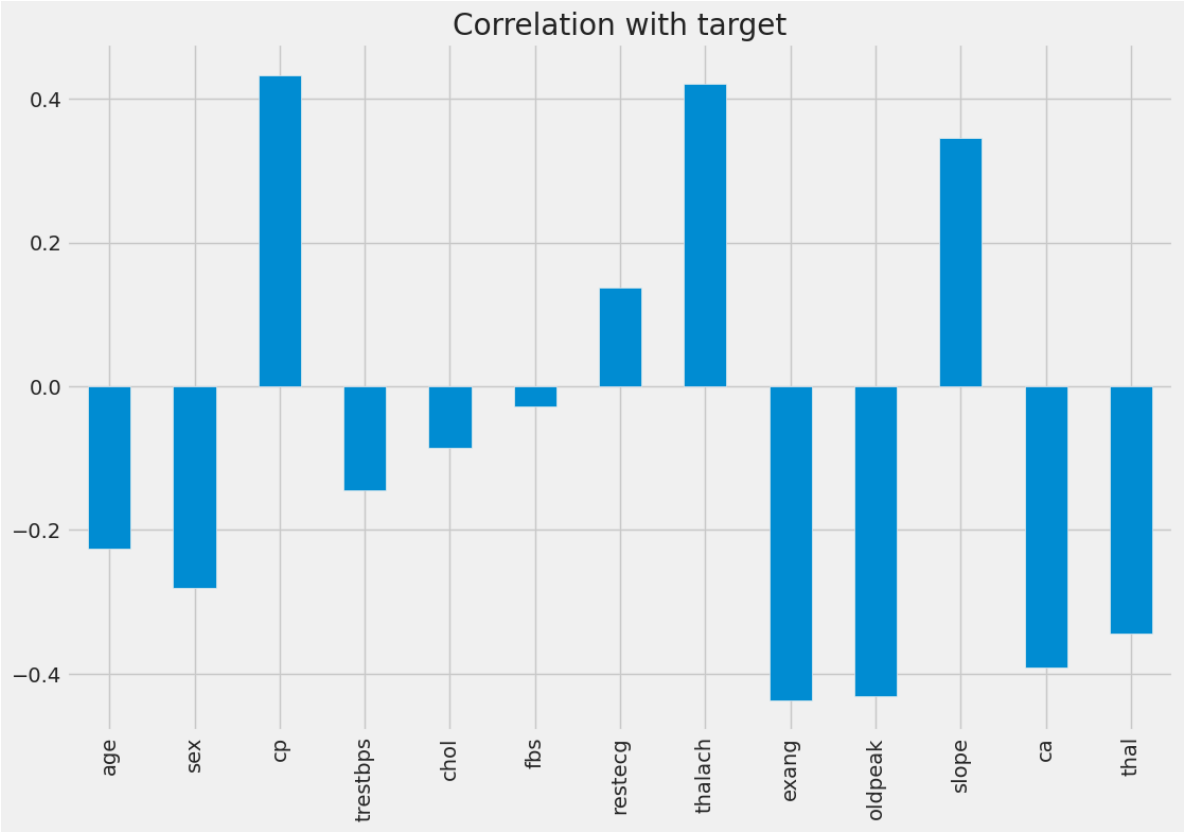
```
(14.5, -0.5)
```




```
df.drop('target', axis=1).corrwith(df.target).plot(kind='bar', grid=True, fi
                                                    title="Correlation with t
```



```
<Axes: title={'center': 'Correlation with target'}>
```



🔍 Observations from Correlation

- **fbs and chol:**
  - fbs and chol are the least correlated with the target variable.
- **Other variables:**
  - All other variables have a significant correlation with the target variable.

Data Processing

After exploring the dataset, we can observe that we need to convert some categorical variables to dummy variables and scale all values before training the machine learning models. So, for this task, I'll use the get_dummies method to create dummy columns for categorical variables:

```
categorical_val.remove('target')
dataset = pd.get_dummies(df, columns = categorical_val)

from sklearn.preprocessing import StandardScaler

s_sc = StandardScaler()
col_to_scale = ['age', 'trestbps', 'chol', 'thalach', 'oldpeak']
dataset[col_to_scale] = s_sc.fit_transform(dataset[col_to_scale])
```

Applying Logistic Regression

Now, I will train a machine learning model for the task of heart disease prediction. I will use the logistic regression algorithm as I mentioned at the beginning of the article.

But before training the model I will first define a helper function for printing the classification report of the performance of the machine learning model:

```
from sklearn.metrics import accuracy_score, confusion_matrix, classification_r

def print_score(clf, X_train, y_train, X_test, y_test, train=True):
    if train:
        pred = clf.predict(X_train)
        clf_report = pd.DataFrame(classification_report(y_train, pred, output_
        print("Train Result:\n================================================
        print(f"Accuracy Score: {accuracy_score(y_train, pred) * 100:.2f}%")
        print("_______________________________________________")
        print(f"CLASSIFICATION REPORT:\n{clf_report}")
        print("_______________________________________________")
        print(f"Confusion Matrix: \n {confusion_matrix(y_train, pred)}\n")

    elif train==False:
        pred = clf.predict(X_test)
        clf_report = pd.DataFrame(classification_report(y_test, pred, output_d
        print("Test Result:\n================================================
        print(f"Accuracy Score: {accuracy_score(y_test, pred) * 100:.2f}%")
        print("_______________________________________________")
        print(f"CLASSIFICATION REPORT:\n{clf_report}")
        print("_______________________________________________")
        print(f"Confusion Matrix: \n {confusion_matrix(y_test, pred)}\n")
```

Now let's split the data into training and test sets. I will split the data into 70% training and 30% testing: